



SIMATS Online Programming Lab

C PROGRAMMING



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CO3 - AT1 - Analytical Problem solving

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QUESTIONS

Q1

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20 marks

**Q2**

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**Q3**

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20 marks

**Q4**

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20 marks

**Q5**

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20 marks



5/5 answered

Q1

CO3 - AT1 - Analytical Problem Solving

20
marks

1. Given an $N \times N$ matrix, Write a C Program to determine whether it is:

- a) Symmetric
- b) Upper triangular
- c) Lower triangular
- d) Diagonal matrix
- e) Identity matrix
- f) Display all applicable classifications.

Your Code

```
1 #include <stdio.h>
2 int main() {
3     int n, i, j;
4     int matrix[10][10];
5     int isSymmetric = 1;
6     int isUpperTriangular = 1;
7     int isLowerTriangular = 1;
8     int isDiagonal = 1;
9     int isIdentity = 1;
10    int classified = 0;
11    scanf("%d", &n);
12    for (i = 0; i < n; i++) {
13        for (j = 0; j < n; j++) {
14            scanf("%d", &matrix[i][j]);
15        }
16    }
17    for (i = 0; i < n; i++) {
```

```
18     for (j = 0; j < n; j++) {
19         if (matrix[i][j] != matrix[j][i])
20             isSymmetric = 0;
21     }
22 }
23
24 for (i = 0; i < n; i++) {
25     for (j = 0; j < i; j++) {
26         if (matrix[i][j] != 0) {
27             isUpperTriangular = 0;
28         }
29     }
30 }
31 for (i = 0; i < n; i++) {
32     for (j = i + 1; j < n; j++) {
33         if (matrix[i][j] != 0) {
34             isLowerTriangular = 0;
35         }
36     }
37 }
38 for (i = 0; i < n; i++) {
39     for (j = 0; j < n; j++) {
40         if (i != j && matrix[i][j] != 0)
41             isDiagonal = 0;
42     }
43 }
44 }
45 for (i = 0; i < n; i++) {
46     for (j = 0; j < n; j++) {
47         if (i == j && matrix[i][j] != 1)
48             isIdentity = 0;
49     }
50     if (i != j && matrix[i][j] != 0)
51         isIdentity = 0;
52 }
53 }
54 }
55 if (isSymmetric) {
56     printf("The matrix is Symmetric.\n");
57     classified = 1;
58 }
59 if (isUpperTriangular) {
60     printf("The matrix is Upper Triangular.\n");
61     classified = 1;
62 }
63 if (isLowerTriangular) {
64     printf("The matrix is Lower Triangular.\n");
65     classified = 1;
66 }
```

```
66     }
67     if (isDiagonal) {
68         printf("The matrix is Diagonal.\n");
69         classified = 1;
70     }
71     if (isIdentity) {
72         printf("The matrix is an Identity");
73         classified = 1;
74     }
75     if (!classified) {
76         printf("The matrix does not satisfy");
77     }
78     return 0;
79 }
```

Input

```
3
1 0 1
1 1 1
0 1 1
```

Output



Run



Save